DAY 3 — Azure OpenAI, APIs & Enterprise Integrations

Module 1: Azure OpenAI Deep Dive

1. GPT Models

Definition

GPT (Generative Pre-trained Transformer) models are Large Language Models (LLMs) capable of:

- Text generation
- Question answering
- Summarization
- Code generation
- Reasoning
- AI chatbot development

Azure OpenAI provides enterprise-grade access to OpenAI models through Microsoft Azure.

Types of GPT Models

Model	Purpose
GPT-4o	Advanced reasoning + multimodal
GPT-4.1	Enterprise-grade complex AI
GPT-4.1-mini	Faster & cheaper
GPT-3.5	Lightweight chatbot workloads
Embedding Models	Semantic search/vector search

Real-Time Example

HR Copilot

User asks:

“Show me leave policy for maternity leave.”

GPT:

- Understands the question

- Searches company policy
- Generates human-like response

Enterprise Use Cases

- AI Helpdesk
- AI HR Assistant
- AI Finance Assistant
- AI Sales Copilot
- AI Knowledge Bots
- AI Email Summarizers

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2. Embeddings

Definition

Embeddings convert text into numerical vectors so AI can understand semantic meaning.

Example:

"Azure OpenAI"
→ [0.234, 0.991, 0.111, ...]

Why Embeddings?

Used for:

- Semantic Search
- Recommendation Engines
- RAG Systems
- Vector Databases
- Similarity Search

Example

User Query

“Vacation leave policy”

AI searches semantically:

- Annual leave
- Paid leave
- Holiday policy

Even if exact words don't match.

Real Enterprise Scenario

AI Policy Search Engine

Employees ask:

"How many sick leaves do I get?"

AI searches embeddings from:

- HR PDFs
- SharePoint
- Dataverse
- SQL DB

Returns relevant answer instantly.

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3. Tokens

Definition

Tokens are chunks of words processed by LLMs.

Example:

Text	Approx Tokens
Hello	1
Azure OpenAI Services	4
One paragraph	100+

Why Tokens Matter?

Tokens affect:

- Pricing
- Performance
- Context window limits
- Latency

Enterprise Example

Large legal documents:

- 50,000 tokens
- AI may need chunking strategy

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4. Temperature

Definition

Temperature controls creativity/randomness in AI responses.

Temperature	Behavior
0.1	Deterministic
0.5	Balanced
1.0	Creative
1.5	Highly random

Example

Prompt

"Write product slogan"

Temp = 0.1

"Reliable AI for Enterprises"

Temp = 1.2

"Unleash Intelligent Innovation Beyond Imagination"

Enterprise Recommendation

Use Case	Temperature
Financial AI	0.1
Medical AI	0.2
Creative Marketing	0.9

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5. Context Windows

Definition

Context window = amount of information AI remembers in one request.

Example

If context window:

128K tokens

AI can process:

- Large PDFs
- Long conversations
- Enterprise knowledge bases

Enterprise Scenario

AI Contract Analyzer

Upload:

- 300-page legal agreement

AI:

- Understands entire document
- Summarizes risks
- Extracts clauses

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6. Function Calling

Definition

Function calling allows GPT to invoke APIs/tools dynamically.

AI decides:

- Which function to call
- What parameters to pass

Example

User

"Book meeting tomorrow at 4 PM"

AI:

- Calls Calendar API
- Creates meeting automatically

Enterprise Example

IT Service Desk AI

User:

"Reset my password"

AI:

- Calls Active Directory API
- Triggers reset workflow

Mini Architecture

User → GPT Model → Function Call → Enterprise API → Response

Module 2: REST API Integration

1. API Concepts

Definition

API (Application Programming Interface) enables communication between systems.

Real Example

Copilot Studio Bot ↔ HR System

Bot asks:

“Show employee leave balance”

API fetches data from backend HR application.

API Types

API Type	Example
REST	Most common
SOAP	Legacy enterprise
GraphQL	Flexible queries

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2. JSON Payloads

Definition

JSON is lightweight structured data format.

Example Request

```
{
  "employeeId": "EMP101",
  "leaveType": "Sick Leave"
}
```

Example Response

```
{
  "balance": 12
}
```

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3. API Authentication

Definition

Authentication secures APIs.

Common Methods

Method	Usage
API Key	Basic APIs
OAuth 2.0	Enterprise apps
JWT Tokens	Secure microservices
Bearer Token	Cloud APIs

Enterprise Example

Microsoft Graph API:

Authorization: Bearer eyJhbGci...

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4. AI API Orchestration

Definition

AI coordinates multiple APIs dynamically.

Scenario

AI Travel Assistant

User:

“Book Bangalore flight and hotel”

AI:

1. Calls airline API
2. Calls hotel API
3. Calls payment API
4. Sends email confirmation

Enterprise Architecture

Copilot Studio
↓
Azure OpenAI

↓
API Orchestrator
↓
ERP | CRM | HRMS | SAP | Salesforce

Module 3: Microsoft Graph API Integration

1. Microsoft Graph Overview

Definition

Microsoft Graph is unified API gateway for Microsoft 365 services.

Services Accessed

- Outlook
- Teams
- SharePoint
- OneDrive
- Users
- Calendar
- Excel
- Planner

Example API

GET https://graph.microsoft.com/v1.0/me

Returns:

- User profile
- Email
- Department

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2. Teams Integration

Example

AI bot inside Microsoft Teams:

- Answer HR queries
- Create tickets
- Schedule meetings
- Notify teams

Real Scenario

IT Support Bot

User:

“VPN not working”

Bot:

- Creates incident
- Notifies IT team
- Updates Teams channel

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3. Outlook Integration

Use Cases

- AI email summarization
- Draft generation
- Smart replies
- Priority classification

Example

AI summarizes:

- 50 unread emails
- Extracts action items

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4. Calendar Integration

Example

AI Assistant:

- Reads availability
- Creates meetings
- Sends reminders

Scenario

Executive Assistant Bot

User:

“Schedule leadership review next Friday”

AI:

- Checks calendars
- Finds free slot
- Sends invite automatically

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5. User Profile Integration

Example

AI fetches:

- Name
- Department
- Manager
- Role
- Location

Enterprise Use Case

Role-based AI responses:

- HR users see HR data
- Finance users see finance data

Module 4: Enterprise Data Integration

1. Dataverse

Definition

Dataverse is Microsoft’s secure business data platform.

Features

- Low-code integration
- Power Platform connectivity
- Security model
- Relational tables

Example

Copilot accesses:

- Employee records
- Tickets
- CRM data

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2. SQL Integration

Example

AI queries enterprise database:

```
SELECT * FROM Employees
WHERE Department='HR'
```

Use Cases

- Sales dashboards
- Employee systems
- ERP integration
- Reporting

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3. External Systems

Examples

System	Integration
SAP	ERP
Salesforce	CRM
ServiceNow	ITSM
Workday	HRMS

Enterprise Scenario

AI assistant integrates:

- SAP purchase orders
- CRM customer details
- Ticketing systems

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4. AI Plugins & Connectors

Definition

Connectors enable AI to interact with external services.

Examples

- SharePoint Connector
- SQL Connector
- ServiceNow Connector
- Dataverse Connector
- REST API Connector

Hands-On Demos

Demo 1 — Call OpenAI APIs from Copilot Studio

Demo Flow

1. Create Copilot Studio agent
2. Create AI action

3. Connect Azure OpenAI endpoint
4. Add API key
5. Send prompt
6. Display AI response

Demo 2 — Connect Microsoft Graph APIs

Demo Steps

1. Register Azure App
2. Configure API permissions
3. Generate client secret
4. Use OAuth authentication
5. Call Graph APIs
6. Retrieve user profile/calendar

Demo 3 — Create Teams-Integrated AI Bot

Demo Steps

1. Create AI bot
2. Publish to Teams
3. Add Teams channel
4. Configure authentication
5. Test enterprise queries

Demo 4 — AI Email Summarization Workflow

Workflow

Outlook Inbox
↓
Graph API
↓
Azure OpenAI
↓
Summarized Email

↓
Teams Notification

Lab 1 — Build AI Meeting Assistant

Objective

Create AI assistant that:

- Reads calendar
- Schedules meetings
- Summarizes emails
- Sends Teams notifications

Features

Feature	**Technology**
Calendar Reading	Graph API
AI Responses	Azure OpenAI
Teams Alerts	Teams API
User Authentication	OAuth

Lab Flow

User Request
↓
Copilot Studio
↓
Azure OpenAI
↓
Graph API
↓
Calendar + Outlook
↓
Meeting Scheduled

Mini Project — Enterprise AI Assistant

Problem Statement

Build enterprise AI assistant for employees.

Features

- AI Q&A chatbot
- Meeting scheduling
- Email summarization
- HR policy search
- SQL integration
- Teams notifications

Technologies

Layer	Technology
Frontend	Copilot Studio
AI Engine	Azure OpenAI
APIs	Microsoft Graph
Database	SQL/Dataverse
Authentication	Azure AD

Expected Outcome

AI assistant capable of:

- Enterprise productivity automation
- Intelligent workflows
- Multi-system integration
- Real-time business assistance

Case Study — AI-Based Executive Assistant Solution

Business Problem

Executives spend:

- 4+ hours daily on emails
- Scheduling conflicts
- Meeting coordination
- Information overload

Proposed AI Solution

AI Executive Assistant:

- Summarizes emails
- Schedules meetings
- Generates meeting notes
- Tracks action items
- Sends reminders

Integrated Systems

System	Purpose
Outlook	Emails
Teams	Collaboration
Calendar	Scheduling
Azure OpenAI	Intelligence
Dataverse	Business data

Benefits

Benefit	Impact
Productivity	Increased
Manual Work	Reduced
Response Time	Faster

Benefit	**Impact**
Meeting Efficiency	Improved

Real Enterprise Scenario

CEO asks:

“Summarize today’s critical emails and schedule meeting with finance head.”

AI:

1. Reads Outlook emails
2. Summarizes priorities
3. Checks calendar
4. Schedules meeting
5. Sends Teams notification